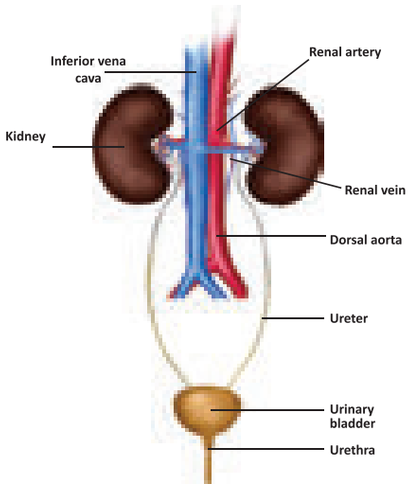
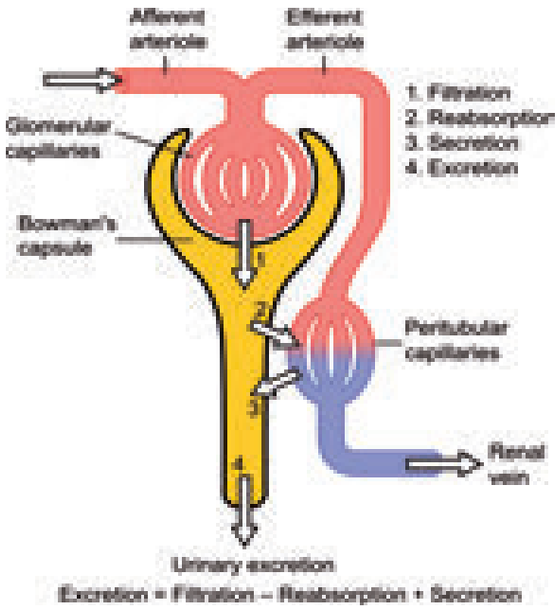
of smooth involuntary muscle which serves as the conduit of urine from the kidneys into the bladder.



Ureters



Urinary System

Bladder

The bladder is the muscular organ that serves as the collection and retention vessel which temporarily holds and retains urine prior to urination. A normal urinary bladder can hold up to 800 milliliters of urine.

Urethra

Urethra transports urine from the bladder to the outside of the body. This image shows (a) a female urethra and (b) a male urethra.

The urethra is the opening through which the urine passes out.

The male gender has two urethral sphincters to control both the passage of urine and sperm and the female gender has only one urethral sphincter to control the flow of urine only.

Physiology Of Urinary System

The functional unit of the urinary system is the nephron. The actual filters are tiny structures called nephron and each kidney is made up of one million nephrons. A nephron is made up of two parts, a cluster of capillaries called glomerulus, and a long fine tube. The nephrons are doing the following functions.

Filtration

Filtration occurs when the circulating blood passes through the kidney where the removal of proteins and other cellular particles. The ultra filtrate becomes urine after reabsorption and secretion.

Re absorption

Reabsorption occurs after filtration, entails the re-entry of some particles and molecules from the ultrafiltrate back into the blood for future use.

NOTE: Anti Diuretic Hormone controls the amount of water in the body and the blood. Excessive water in the blood causes high blood pressure and fluid overload, and a low amount of water in the blood causes low blood pressure and dehydration. Anti diuretic hormone also decreases the amount of aldosterone in the body.

Secretion

Secretion is the opposite of reabsorption which entails the movement of wastes and

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